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MIL-STD-1864A

14 February 1991

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6 JUN 1983

MILITARY STANDARD

FIBER OPTIC SYMBOLS

AMSC N/A

FSC 60GP

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DEPARTMENT OF DEFENSE

Washington, DC 20301

FIBER OPTIC SYMBOLS

MIL-STD-1864A

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2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Communications-Electronics Command, DRSEL-ED-TO, Fort Monmouth NJ 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3. DEFINITIONS

3.1 General. Definitions shall be in accordance with IEEE-STD-812.

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1. SCOPE

1.1 Purpose. This standard gives a list of symbols for fiber optic parts for use on military drawings and wherever symbols for fiber optic parts are required.

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2. REFERENCED DOCUMENTS

2.1 Other publications. The following documents form a part of this standard to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Institute of Electrical and Electronics Engineers (IEEE)

IEEE-STD-812 - IEEE Standard Definition of Terms Relating to Fiber Optics.

(Application for copies should be addressed to the IEEE, 345 East 47th Street, New York, NY 10017.)

American National Standards Institute (ANSI)

ANSI Y32.2 (IEEE315) - Graphic Symbols for Electrical and Electronics Diagrams (including reference designation class designation letters)

IEC 617 - Recommended Graphical Symbols.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

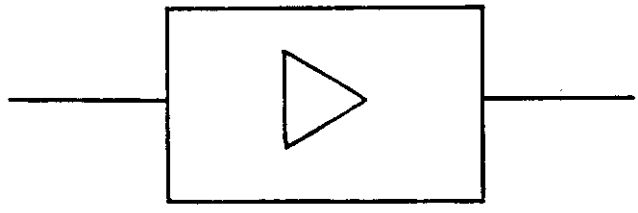
2.2 Order of precedence. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.

2.3 Source of documents. Copies of industry association documents should be obtained from the sponsor. Copies of all other listed documents should be obtained from the contracting activity or as directed by the contracting officer.

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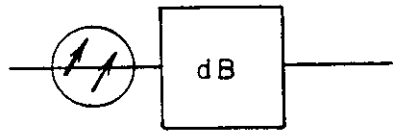
5. DETAILED SYMBOL LIST

5.1 Amplifier, optical, electrical.

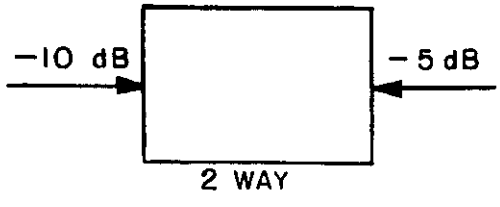


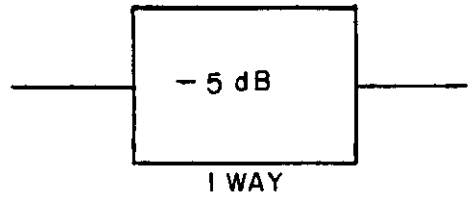
NOTE: Indicate the dB level.

5.2 Attenuator, optical.



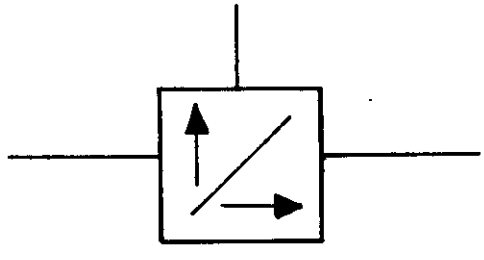
5.2.1 Attenuator, optical, electrical.





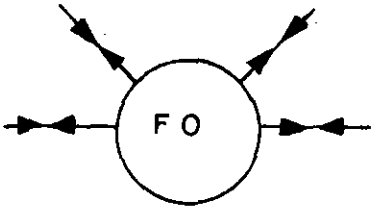
NOTES:
1. Arrows on cable optional.
2. Directional value specified, optional (db loss).
3. If dB value specified for two way (directional), the two attenuation values will be outside of the block with an arrow pointing in the direction of the optical beam.
4. In case of one way, the value (dB loss) will be inside of the block.

5.3 Beam splitter.



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5.4 Coupler.



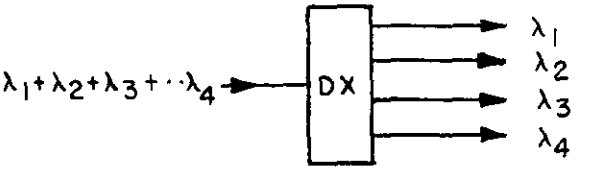
Use standard established graphic symbols of ANSI Y32.2 for coupler in conjuction with the following:

NOTES:
1. "Tp" for transmissive passive.
2. "Ta" for transmissive active.
3. "Rp" for reflective passive.

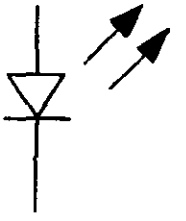
5.5 Delay device, optical.



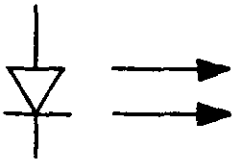
5.6 Demultiplexer, wavelength.



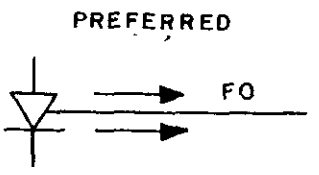
5.7 Diode, laser, (ILD).

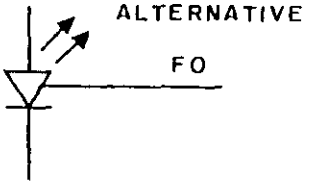


5.8 Diode, light emitting, (LED).



5.8.1 Pigtailed.

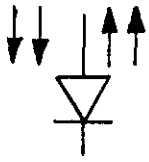




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5.9 Diode, light emitting-photodetector (DETEM).

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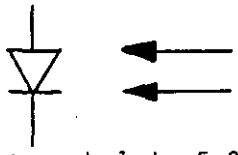


ALTERNATIVE

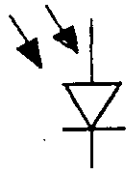


5.10 Diode, photodetector. The same symbol is used for avalanche photo diodes (APD) and pin diodes.

PREFERRED

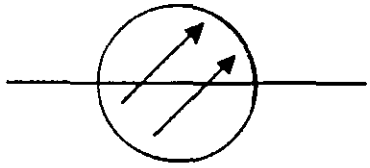


ALTERNATIVE



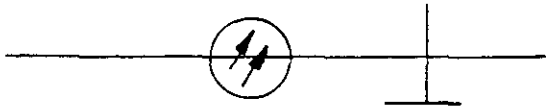
5.10.1 Pigtailed. Use the symbol in 5.8.1 except reverse direction of the arrows.

5.11 Fibers.

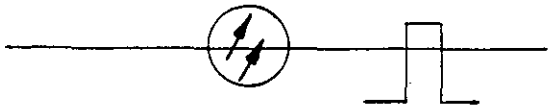


NOTE: For international or NATO application, the following symbol shall be used to indicate optical fiber in accordance with International Electrotechnical Commission publication IEC 617.

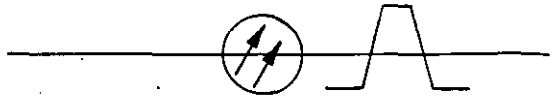
5.11.1 Single mode stepped index.



5.11.2 Multimode stepped index.



5.11.3 Graded index.

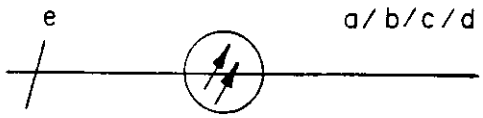


NOTE: The fiber index identifiers for each type of fiber construction should be placed adjacent to the symbol element denoting an optical fiber as shown to avoid confusion with signal waveforms.

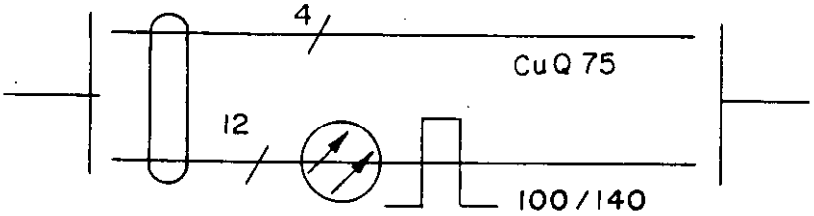
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5.11.4 Supplementary information.

a = core
b = cladding
c = first buffer
d = jacketing
e = number of fibers



5.11.5 Composite cable.



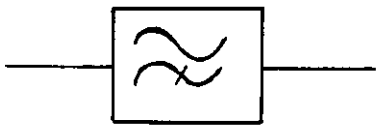
NOTE: Example of a composite cable containing both copper conductors and optical fibers.

5.11.6 Permanent joint.

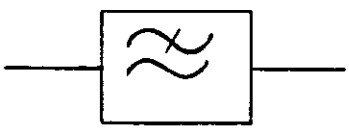


5.12 Filters. Above each symbol, the limiting wavelengths shall be provided (in nm).

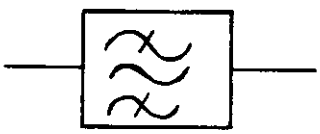
5.12.1 Long wavelength pass.



5.12.2 Short wavelength pass.

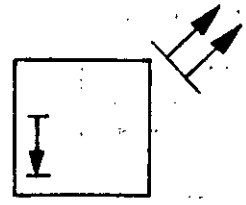


5.12.3 Band pass.

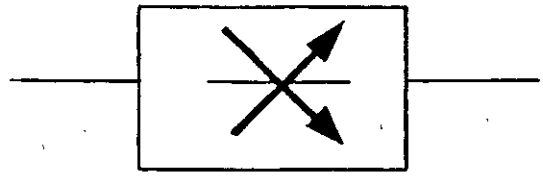


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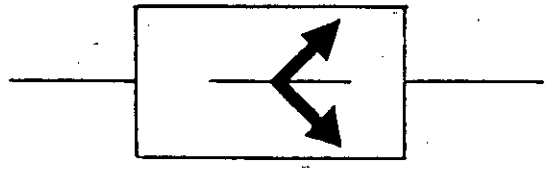
5.13 Laser (except laser diode).



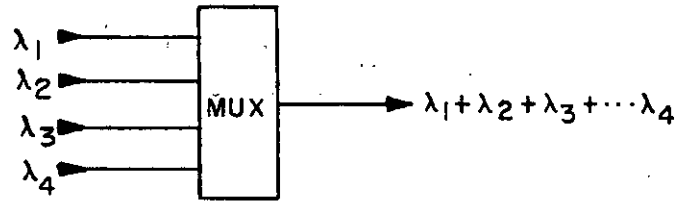
5.14 Mode mixer.



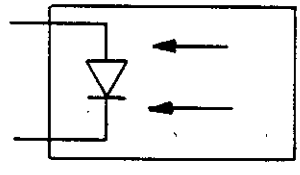
5.15 Mode stripper.



5.16 Multiplexer, wavelength.

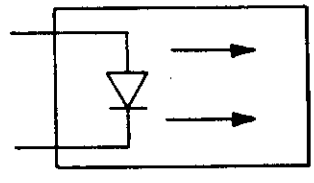


5.17 Receivers.



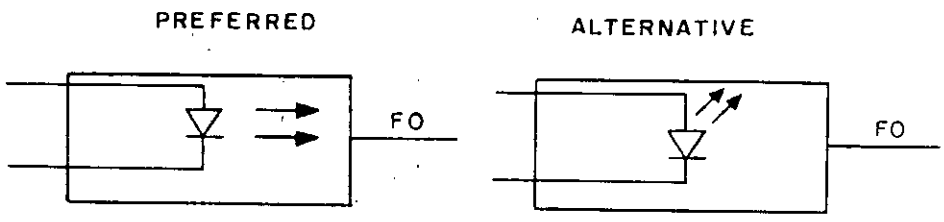
5.18 Repeater. See amplifier, optical.

5.19 Transmitter.

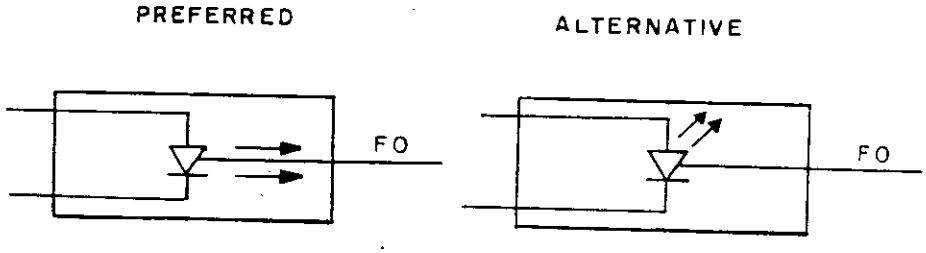


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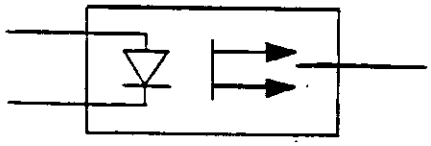
5.19.1 Without pigtail source.



5.19.2 With pigtail source.

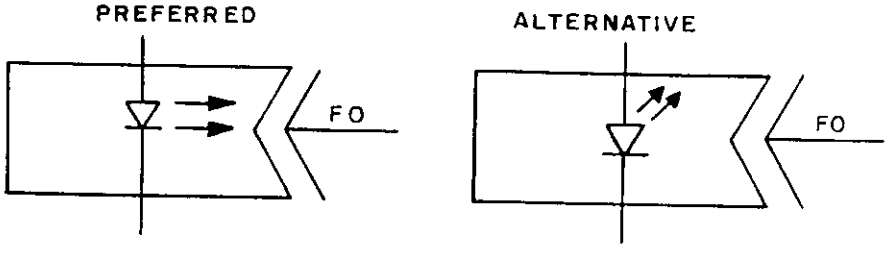


5.19.3 Coherent guided with laser diode.



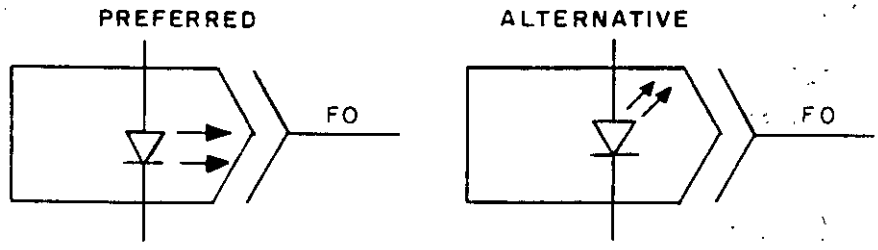
5.20 Connectors. Symbol for connectors shall be that shown in ANSI Y32.2 with two exceptions shown in 5.19.1 and 5.19.2.

5.20.1 Receptacle with optical source. For optical detectors, reverse direction of arrows.

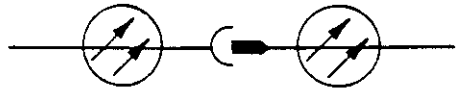


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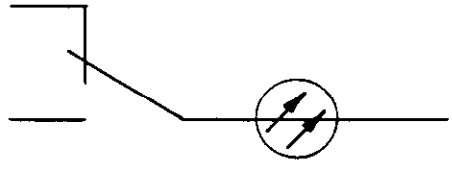
5.20.2 Plug with optical source. For optical detectors, reverse direction of arrows.



5.20.3 Connection female-male.



5.20.4 Change-over contact.



6. NOTES

6.1 Intended use. The purpose of this standard is to standardize the symbols used for fiber optic parts on military drawings, DOD specifications and standards, wherever symbols for fiber optics are required.

6.2 Subject term (key word) listing.

Graded index
Laser
Mode
Multiplex
Receiver
Step-index
Transmitter

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-STD-1864A	2. DOCUMENT DATE (YYMMDD) 14 February 1991

3. DOCUMENT TITLE

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Commander U.S. Army Communications-Electronics Command	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code) ATTN: AMSEL-ED-TM Fort Monmouth, NJ 07703-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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